

Md Rahman

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EDUCATION

City College of New York, Grove School of Engineering

New York, NY

B.E. in Mechanical Engineering

Expected May 2029

- GPA: 4.00/4.00 — Dean's List — S. Jay Levy Fellowship (2026–2027) — Celia Portnoy Memorial Scholarship (Spring 2026)
- MIST National Tournament: 1st Place Extemporaneous Speaking, 3rd Place Debate (2026)
- Coursework: ME 24600 Engineering Mechanics I/Statics (A+), Calculus III (A), ME 14500 Computer-Aided Drafting (A+), University Physics I & II (A+)

Stuyvesant High School

New York, NY

High School Diploma

2021 – 2025

TECHNICAL SKILLS

CAD & Fabrication: SolidWorks, Onshape, FDM 3D Printing (Bambu Lab P1P), Laser Cutting, Machining, STL Preparation

Analysis & Programming: MATLAB, Python (NumPy, Matplotlib, Data Visualization), Sensor Calibration

Controls & Electronics: Arduino (Embedded C), PID Control Loop Tuning, IMU/Encoder Integration

Hardware & Testing: Test Fixture Design, Bench Load Testing, Digital Calipers, Rapid Prototyping

ENGINEERING RESEARCH & PROJECTS

BioWearLab – Grove School of Engineering

New York, NY

Undergraduate Research Assistant, Mechanical Engineering

2026 – Present

- Engineered BCC lattice metamaterials in SolidWorks for additively manufactured TPU–graphene pressure/force sensors, optimizing unit-cell geometry across 8+ iterations to target gait-force ranges of 0–50 N and 0–100 N with less than 5% deviation from target stiffness.
- Designed and manufactured 5+ custom test fixtures, compression mounts, and loading rigs on a Bambu Lab P1P (+/-0.2 mm dimensional tolerance verified via digital calipers), enabling repeatable sensor characterization under compression, shear, and cyclic loading conditions.
- Architected automated data processing pipelines in Python and MATLAB to analyze 500+ data points per test cycle; generated performance curves, force-displacement plots, and summary tables that reduced weekly reporting time by approximately 40%.

Competition Robot – StuyPulse 694

New York, NY

Mechanical Lead

2024 – 2025

- Designed a custom drivetrain and two-stage gearbox in SolidWorks/Onshape for a 120 lb competition robot; achieved 12% mass reduction (4.8 lbs saved) through plate pocketing, ribbed geometry, and topology-informed material removal; verified stiffness via hand calculations and bench load testing.
- Integrated FDM-printed prototype assemblies (intake, manipulator, mounting brackets) with PID control loops using encoder and IMU feedback; tuned gain parameters over 15+ test iterations to achieve autonomous path accuracy of +/-3 cm across a 5 m traverse.

LEADERSHIP & OPERATIONS

Khan's Tutorial

Queens, NY

Branch Manager

2024 – Present

- Oversee operations, student enrollment, and instructional teams across 6 branch locations in Queens and Brooklyn, serving 90+ students and 8 instructors.
- Closed 10+ enrollments totaling \$15,000+ in revenue via Pipedrive CRM-tracked outreach; reduced missed sessions by 20% through systematic attendance tracking and automated reminders via Dialpad and Stripe.

MIST New York

New York, NY

Operations Lead

2024 – Present

- Directed comprehensive day-of logistics and crisis management for a 1,150-attendee tournament across Adelphi University, coordinating 80+ volunteers via two-way radio with real-time room reassignments and AV troubleshooting.
- Executed 30+ complex competition events at 8% under budget through bulk procurement and vendor negotiation, while serving as lead MC for an 800-person awards ceremony at City Tech Theater.